

# William Phong

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## EDUCATION

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### San Diego State University

Aug 2025 – May 2027

*Master of Science in Computer Science*

### California State University San Marcos

May 2024

*Bachelor of Science in Computer Science*

- **Honors:** Cum Laude, Deans List Spring 2023 & 2024

## EXPERIENCE

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### Graduate Research Assistant

Sep 2025 – Present

*SDSU Research Foundation*

*San Diego, CA*

- Incorporating over 200 GB of NOAA datasets for 3D-visualization and statistical analysis
- Building responsive frontend using Next.js, React, TypeScript, Tailwind CSS, and Cesium.js
- Architected data pipeline through ZARR/NetCDF files, PostgreSQL, Docker, Python, FastAPI, and Drizzle
- Created interactive data visualizations, including time series analyses, trends, histogram, periodogram, and spectrogram

### Graduate Research Intern

Sep 2025 – Present

*Pisces Lab @ SDSU*

*San Diego, CA*

- Building an agentic digital-twin system for smart city traffic using CARLA and Ruth simulators
- Integrated Google Gemini for natural language translation, autonomous entity control, and inferencing
- Constructed front-end displaying simulation data using Next.js/React, TypeScript, and Tailwind CSS

### Data Science Intern

Jun 2025 – Aug 2025

*Tensor Therapeutics*

*San Diego, CA*

- Accelerated drug discovery processes using Boltz-2 to predict for RNA structures and molecular interactions
- Deployed a scalable variant calling pipeline using Nextflow Sarek and Docker for genomic data analysis
- Configured, managed, and scaled virtual machines on Azure and Lambda.ai, implementing secure networking

## PROJECTS

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### VR Earth Orbit Simulation

- Collaborated with Dr. Kostadinov to implement research on Earth's orbit, Milankovitch cycles, and insolation
- Developed model in C# to generate mathematically accurate real-time simulations and data calculations
- Provides an intuitive, hands-on VR environment for teaching complex astronomical concepts

### Spotify Day List Word Cloud

- Generate word clouds over time based on Spotify day list titles, displays words weighted by occurrence
- Utilizes Spotify API for OAuth 2.0 code flow to obtain user information
- Written with Python, Matplotlib, and uses AWS RDS Postgres DB to store JSON data

### Formula1 Result Prediction Algorithm

- Implements machine learning algorithms in Python to predict results based on qualifying data
- Data is imported/exported to a Postgres database and visualized with Matplotlib

### Student Portal Application

- Interviewed faculty admin and students of CSUSM to determine a customer story and requirements
- Conducted agile and scrum scheduling, wrote SRS and project documentation
- Created application with Java and Android Studio with MySQL database and SHA256 hashing

## SKILLS

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**Languages:** C++, C#, Python, Java, SQL, TypeScript, JavaScript, HTML, CSS

**Frameworks:** React, Next.js, Node.js, Matplotlib, Pandas, TensorFlow, jQuery, FastAPI, Flask, Django, JUnit

**Developer Tools:** Git, Docker, Amazon Web Services, Google Cloud, Microsoft Azure, Cloudflare Pages